

Analysis of Aadhaar Enrolment Patterns in India

A Data-Driven Study Using UIDAI Enrolment Data

-UIDAI Hackathon 2026

Team Name: DataSynth

Team Members: BHUVAN M S

PURVI S KIRAN

Institution: Nitte Meenakshi Institute Of Technology, Bangalore

Date

January 2026

This report presents a comprehensive analysis of Aadhaar enrolment data published by UIDAI, focusing on temporal trends, regional distribution, and demographic patterns. The study aims to derive actionable insights that support operational planning and evidence based decision making.

Scope of Study

This study covers Aadhaar enrolment data across all Indian states and union territories over the available time period. The analysis focuses on identifying temporal trends, regional disparities, and demographic patterns in enrolment to support operational and policy level insights for UIDAI.

STATEMENT & APPROACH

Problem Statement

Aadhaar plays a critical role in enabling identity-based service delivery across India. Understanding enrolment patterns across time, geography, and demographic groups is essential for improving operational efficiency, ensuring equitable coverage, and supporting evidence-based decision-making.

This project aims to analyze Aadhaar enrolment data provided by UIDAI to identify temporal trends, regional variations, demographic contributions, and unusual enrolment patterns.

Approach

The analysis follows a structured data analytics approach:

1. Collection of official UIDAI Aadhaar enrolment data
2. Data cleaning and preprocessing to ensure consistency and reliability
3. Exploratory data analysis across:
 - Time
 - States
 - Age groups
4. Identification of key insights and anomaly patterns
5. Translation of insights into actionable recommendations

The objective of this project is to analyze Aadhaar enrolment data provided by UIDAI to uncover meaningful patterns related to:

- When enrolments occur,
- Where enrolments are concentrated,
- Which demographic groups contribute most to enrolment activity.

The analysis aims to transform raw enrolment data into insights that can support administrative decision-making and improve enrolment strategies.

DATASETS USED

Dataset Source

The analysis uses the official **Aadhaar Enrolment Dataset** published by UIDAI and made available through the data.gov.in platform. The dataset represents authentic, government-verified enrolment records.

Dataset Description

The dataset contains daily enrolment information across Indian states and districts, with enrolment counts segmented by age groups. It enables both granular and aggregated analysis of Aadhaar registrations.

Columns Used

- date: Date of enrolment record
- state: State or Union Territory
- district: District name
- pincode: Geographic pincode
- age_0_5: Enrolments for children aged 0–5
- age_5_17: Enrolments for individuals aged 5–17
- age_18_greater: Enrolments for adults aged 18 and above

Derived Fields

To support analysis, additional variables were derived:

- Total enrolment per record
- Year and month extracted from date values

DATA CLEANING & PREPROCESSING

Raw administrative datasets such as Aadhaar enrolment records are not directly suitable for analysis. They often contain inconsistent date formats, mixed data types, missing or invalid entries, and redundant information. Therefore, a structured data cleaning and preprocessing pipeline was implemented to ensure that all subsequent analysis is accurate, consistent, and reproducible.

The UIDAI Aadhaar enrolment dataset was processed through the following systematic steps.

1. **Data Standardization**

All date values were converted into a consistent datetime format, and records with invalid or unparseable dates were removed to ensure accurate time-based analysis.

2. **Data Type Validation**

Age-group enrolment fields and pincode values were converted to numeric data types to avoid calculation and aggregation errors.

3. **Feature Engineering**

A new column, **Total Enrolment**, was created by summing the enrolments across all age groups, enabling simpler and more meaningful analysis.

4. **Consistency Checks**

State and district names were standardized to prevent duplication and ensure accurate grouping during state-wise and regional analysis.

5. **Final Data Verification**

The cleaned dataset was checked for missing values, data type consistency, and logical validity before being saved as the final analysis-ready dataset.

Tools & Technologies

- Python for data processing
- Pandas for data manipulation
- Matplotlib for visualization

TIME-BASED ANALYSIS (MONTHLY TRENDS)

Objective

The purpose of time-based analysis is to understand how Aadhaar enrolments fluctuate over time and to identify periods of high and low activity.

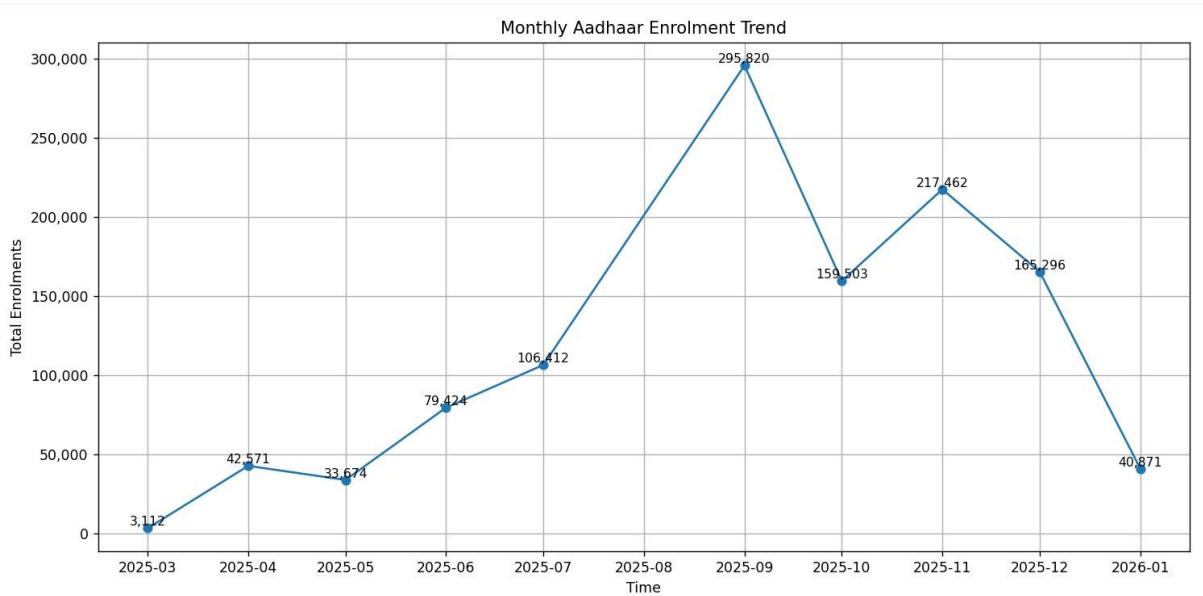
Method

Daily enrolment data was aggregated into monthly totals to smooth short-term fluctuations and highlight broader patterns.

```
PS C:\Users\bhuva\OneDrive\Documents\UIDAI hackathon> python time_based_analysis.py

Monthly Enrolment Summary:
year_month  total_enrolment
0 2025-03-01      3112
1 2025-04-01     42571
2 2025-05-01     33674
3 2025-06-01     79424
4 2025-07-01    106412
5 2025-09-01    295820
6 2025-10-01    159503
7 2025-11-01    217462
8 2025-12-01    165296
9 2026-01-01     40871

Yearly Enrolment Summary:
year  total_enrolment
0 2025    1103274
1 2026     40871
```



Key Observations

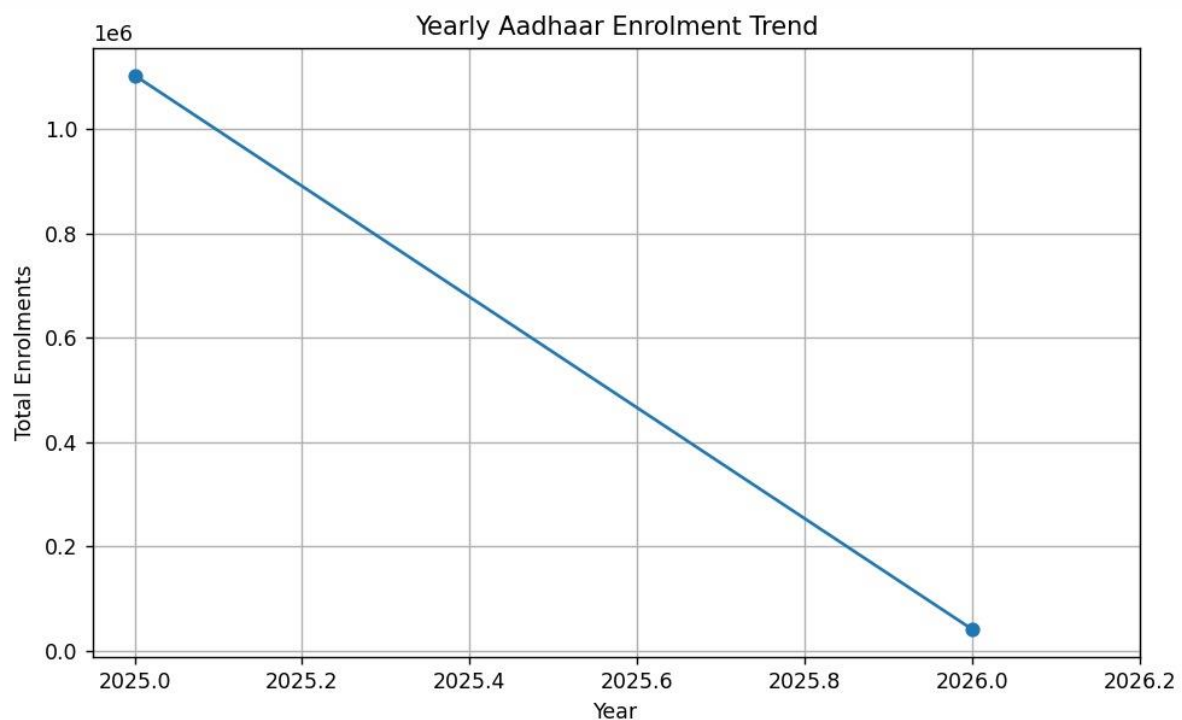
- Aadhaar enrolment exhibits noticeable variation across months
- Certain months record significantly higher enrolment volumes
- Lower values at the beginning and end of the timeline are influenced by partial data coverage

This suggests that enrolment activity is influenced by operational and temporal factors.

TIME-BASED ANALYSIS (YEARLY TRENDS)

Objective

Yearly aggregation provides a high-level perspective on enrolment trends.



Key Observations

- The year 2025 accounts for a larger share of enrolments compared to 2026
- This difference is primarily due to incomplete data availability for 2026
- The trend should therefore be interpreted with caution

STATE-WISE ANALYSIS

Objective

State-wise analysis helps identify regional disparities in Aadhaar enrolment.

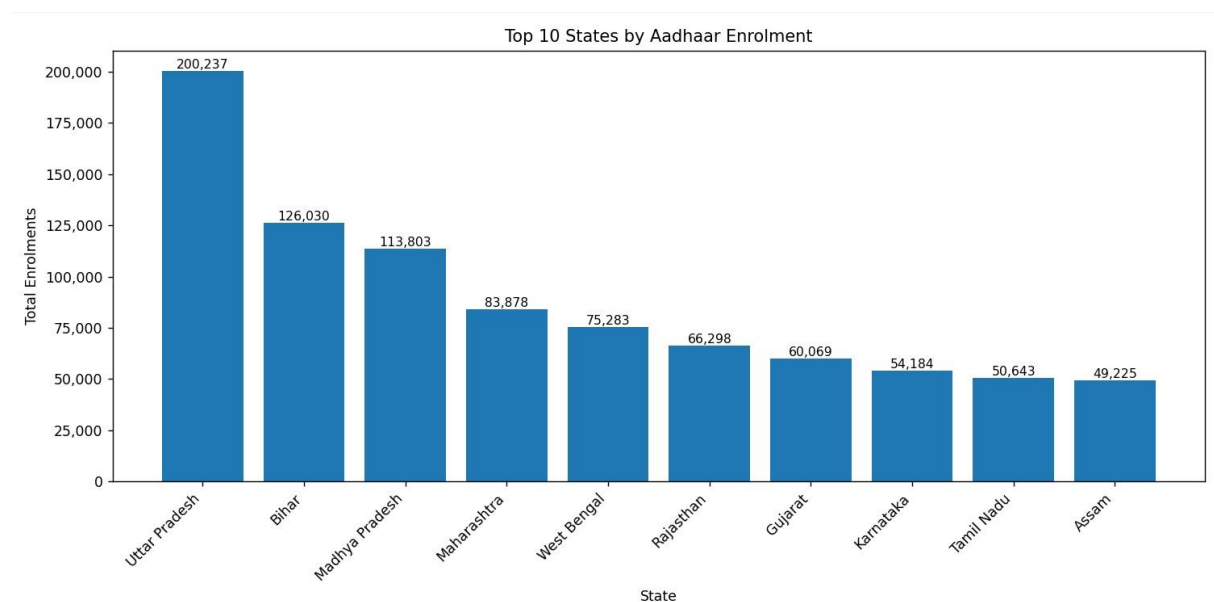
Top-Performing States

```
PS C:\Users\bhuva\OneDrive\Documents\UIDAI hackathon> python state_wise_analysis.py

State-wise Enrolment Summary (Top 10):
state total_enrolment
0 Uttar Pradesh 200237
1 Bihar 126030
2 Madhya Pradesh 113803
3 Maharashtra 83878
4 West Bengal 75283
5 Rajasthan 66298
6 Gujarat 60069
7 Karnataka 54184
8 Tamil Nadu 50643
9 Assam 49225

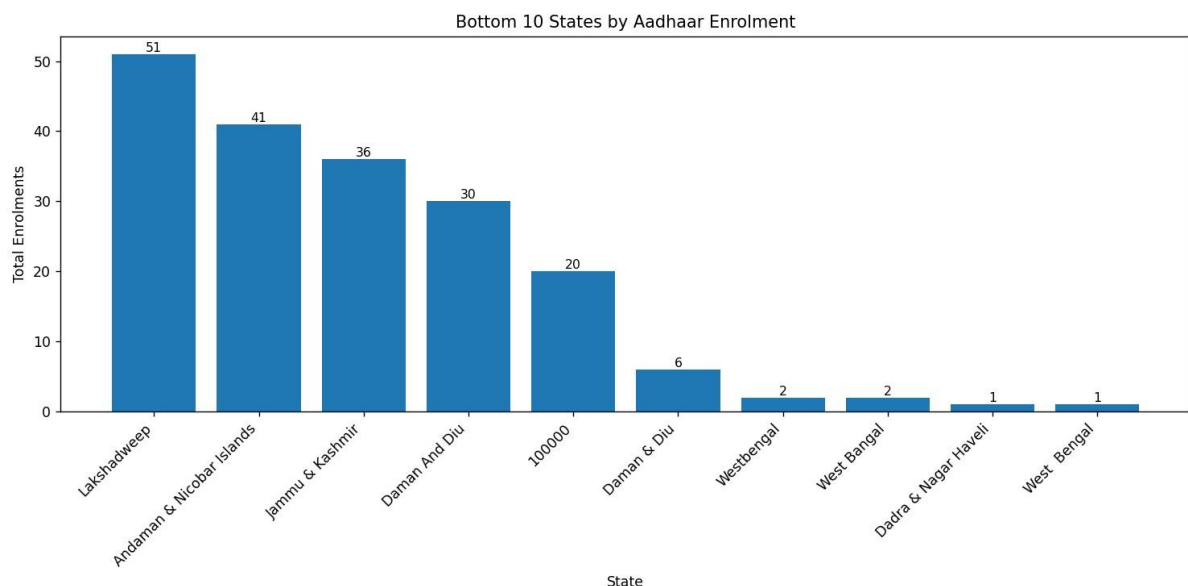
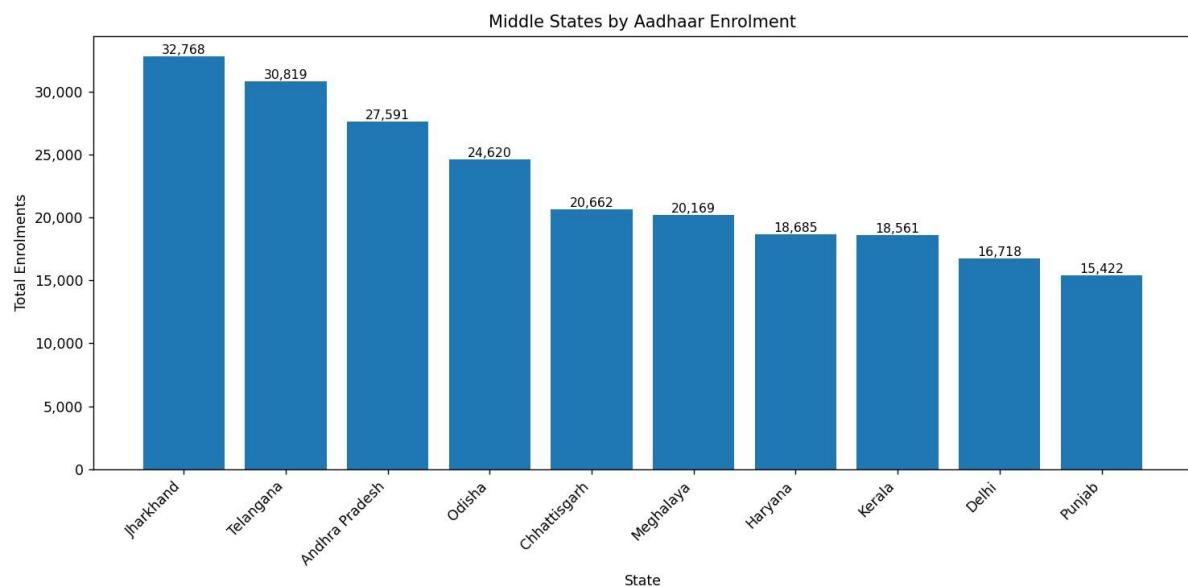
Middle States Enrolment Summary:
state total_enrolment
10 Jharkhand 32768
11 Telangana 30819
12 Andhra Pradesh 27501
13 Odisha 24620
14 Chhattisgarh 20662
15 Meghalaya 20169
16 Haryana 18685
17 Kerala 18561
18 Delhi 16718
19 Punjab 15422

State-wise Enrolment Summary (Bottom 10):
state total_enrolment
39 Lakshadweep 51
40 Andaman & Nicobar Islands 41
41 Jammu & Kashmir 36
42 Daman And Diu 30
43 100000 20
44 Daman & Diu 6
45 Westbengal 2
46 West Bangal 2
47 Dadra & Nagar Haveli 1
```



- A small number of states contribute a disproportionately large share of enrolments
- These states represent regions with high population density and enrolment demand

Middle & Bottom States



Most states fall within a moderate enrolment range

Some states consistently record lower enrolments, indicating potential accessibility or awareness gaps

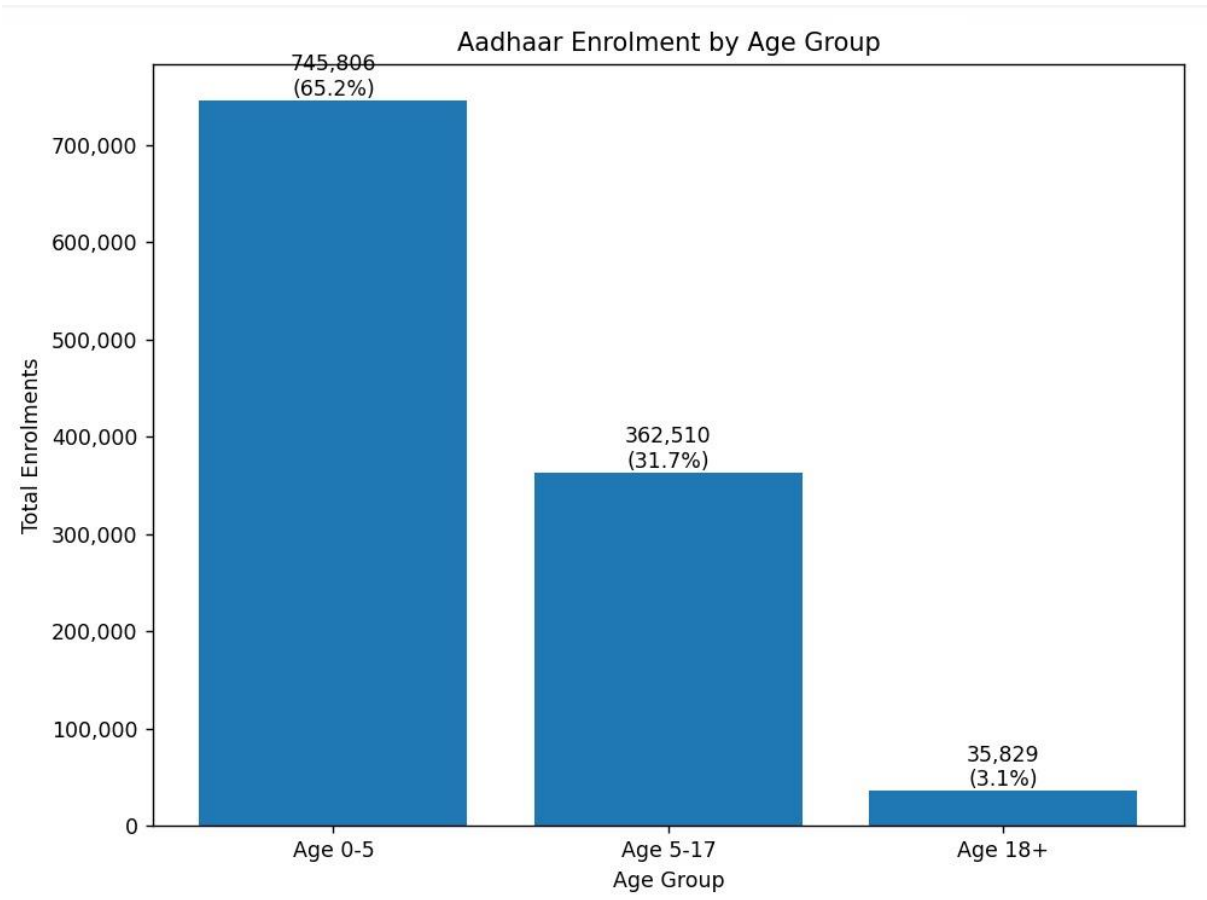
AGE-GROUP ANALYSIS

Objective

This analysis examines how Aadhaar enrolment varies across age groups.

```
PS C:\Users\bhuva\OneDrive\Documents\UIDAI hackathon> python age_group_analysis.py

Age Group Enrolment Summary:
Age Group  Total Enrolment  Percentage
0  Age 0-5      745806    65.184570
1  Age 5-17     362510    31.683921
2  Age 18+       35829     3.131509
```



Key Observations

- Adult enrolments (18+) form the largest share of total registrations
- Child enrolments (0–17 years) are comparatively lower
- This highlights opportunities to strengthen enrolment initiatives for younger populations

KEY INSIGHTS & ANOMALIES

Key Insights

- Aadhaar enrolment varies significantly across time and regions
- Enrolment is concentrated among a limited number of states
- Adults are the primary contributors to enrolment activity

```
PS C:\Users\bhuva\OneDrive\Documents\UIDAI hackathon> python insights_anomalies.py

KEY INSIGHTS
-----
Total Aadhaar enrolments analysed: 1,144,145
Adult (18+) enrolments dominate with 3.1% share
Child enrolments (0-17) contribute 96.9% of total
Top 5 states contribute 52.4% of total enrolments

HIGH ENROLMENT MONTHS (Anomalies)
-----
  year_month  total_enrolment
5 2025-09-01      295820
7 2025-11-01      217462

LOW ENROLMENT MONTHS (Anomalies)
-----
  year_month  total_enrolment
0 2025-03-01       3112
```

Anomaly Observations

- Certain months show unusually high enrolment compared to the overall average
- Periods of low enrolment may reflect seasonal or operational factors

FINAL CONCLUSIONS

Conclusions

- Aadhaar enrolment patterns are influenced by temporal, regional, and demographic factors
- Significant regional disparities exist across states
- Adult enrolments dominate overall registration figures
- Time-based fluctuations indicate the impact of enrolment drives and administrative actions

RECOMMENDATIONS & FUTURE SCOPE

Recommendations

1. Optimize Resource Allocation During Peak Periods

Monthly enrolment trends can be used to allocate staff, infrastructure, and enrolment kits more efficiently during high-demand months to reduce waiting time and service bottlenecks.

2. Improve Coverage in Low-Enrolment States

States showing consistently low enrolment should be targeted with awareness drives, mobile enrolment units, and partnerships with local administration to improve Aadhaar penetration.

3. Strengthen Child Aadhaar Programs

Since enrolments in the 0–17 age group are comparatively lower, UIDAI can focus on school-based and hospital-based Aadhaar registration programs to improve early-age coverage.

4. Use Data for Operational Planning

Regular analysis of enrolment data should be used to plan future enrolment campaigns, location of centres, and staff deployment based on demand patterns.

5. Develop Real-Time Monitoring Systems

Integrating enrolment data into dashboards can help UIDAI track performance, identify delays, and take corrective actions more quickly.

Future Scope

1. Time-Series Forecasting of Enrolments

Historical Aadhaar enrolment data can be used to build forecasting models that predict future enrolment demand and help UIDAI plan infrastructure and staffing.

2. District-Level and Pincode-Level Analysis

Extending the analysis to district and pincode levels can identify localised gaps in Aadhaar coverage and enable more targeted interventions.

3. Integration with Socio-Economic Indicators

Combining enrolment data with population, literacy, income, and internet-penetration data can provide deeper insight into factors affecting Aadhaar adoption.

4. Automated Anomaly Detection Systems

Machine learning techniques can be applied to automatically detect unusual spikes or drops in enrolment, helping administrators respond quickly to issues.

5. Real-Time Enrolment Dashboards

Developing interactive dashboards can allow UIDAI officials to monitor enrolment trends, regional performance, and age-group distribution in real time.

6. Analysis of Aadhaar Update and Correction Records

Including data on Aadhaar updates and corrections can help understand user behaviour after initial registration and identify areas needing better data quality.

7. Evaluation of Enrolment Campaign Effectiveness

Future studies can compare enrolment trends before and after awareness campaigns to measure their impact and improve future strategies.

8. Incorporation of Geographic and Infrastructure Data

Mapping enrolment centres, connectivity, and population density alongside enrolment data can support better planning of new centres and outreach programs.